Here is each problem from the problem statement, matched to how the solution handles it.

Problems in the background section

#	Problem stated	How the solution solves it	Where it happens
1	Records are fragmented across manual correspondence and multiple tracking mechanisms	One central portal holds every institution's FDRs, documents, status and history, so Excel sheets, emails and files are no longer needed	Central repository; Institution portal; Scrutiny workspace
2	Hard to keep an accurate, up-to-date view of deposit status	Every FDR has one status that can only move along allowed steps (Submitted, Under Scrutiny, Active, Expiring Soon, Renewed, Released, Closed), so it always shows the true stage	Status state machine; Dashboard
3	Pending actions and time-sensitive requirements are hard to see	The "Needs attention today" list ranks cases by risk and urgency, and KPI cards show pending verifications, overdue actions and open discrepancies	Command-centre dashboard
4	Submitted financial instruments are difficult to verify	The AI reads the FDR scan and rules check amount, dates, holder name and lien. The officer then runs bank verification and gets Match, Mismatch or Unable to verify	Document AI; Bank verification adapter; Scrutiny officer
5	Institution-submitted data must be told apart from bank-confirmed data	Every field carries a badge: Institution-declared, AI-extracted, Bank-confirmed or Officer-manual. A side-by-side comparison table shows the differences	Instrument detail, Verification tab
6	Validity and maturity dates are hard to monitor	A scheduler checks every date each day, moves FDRs to Expiring Soon or Matured automatically, and shows a 12-month maturity chart	Monitoring and alerts engine
7	Renewal or replacement requirements are hard to track	New FDRs are linked to the old one in a version chain. A gap alert fires if the old FDR lapses with no replacement	Renewal and replacement tracking
8	Release and refund requests are hard to record and track	A request goes through an automatic eligibility check, then Scrutiny, Accounts and Authority approval, then a release order and closure, with remarks and a timer at each stage	Release and refund workflow
9	Documents must stay available for regulatory verification	Documents are stored securely with version history and a SHA-256 fingerprint, and can be previewed and retrieved from the case at any time	Document management

10	Repetitive administrative work	The AI pre-fills forms, letters and reminders are drafted by the AI (officer confirms), and reports can be exported in one click	AI drafting; Reports centre
11	Records may be inaccurate or incomplete	Form validation (IFSC, dates, amounts, file type), AI confidence scores, and a missing-documents report catch gaps early	Submission wizard; Discrepancy detection
12	Concerned officials and institutions need timely alerts	Reminders go out at 90, 60, 30, 15, 7 and 1 days before expiry, and again after the deadline, by app, email and SMS (email and SMS are simulated in the prototype)	Alerts engine
13	Tracking must be transparent across the whole lifecycle	Each case has a full timeline, and institutions can see their own status and next action	Timeline tab; Institution portal
14	Must follow AICTE requirements and authorized financial procedures	Required amount and document checklists are configurable by the admin. Release is blocked automatically if conditions are not met, and only the competent authority can approve	Admin rules; Release workflow; RBAC

Problems in the detailed description and expected solution

#	Problem stated	How the solution solves it	Where it happens
15	Incomplete records and discrepancies need to be flagged	Hybrid rule and AI checks flag low amount, wrong holder name, lien not marked to AICTE, and duplicate FDR numbers, each with a severity and a plain-language reason	Discrepancy register
16	Overdue and unresolved cases need follow-up	Overdue cases show in red with a timer. If no one acts, the case moves up the ladder from institution nodal officer to the competent authority	Escalation matrix
17	Need an overview of active, expiring, matured, renewed, released and pending instruments	Dashboard cards and charts cover each of these. Every card opens the filtered list behind it	Dashboard
18	Officials need to search records, check document history and record observations	Search and filters by institution, state, bank, status and amount, plus a documents tab with versions and an observations tab with replies	Scrutiny workspace
19	Reports for pending verifications, upcoming maturities, missing documents and cases needing intervention	Ready reports for each of these, plus discrepancy and release registers, exportable to CSV and PDF	Reports centre

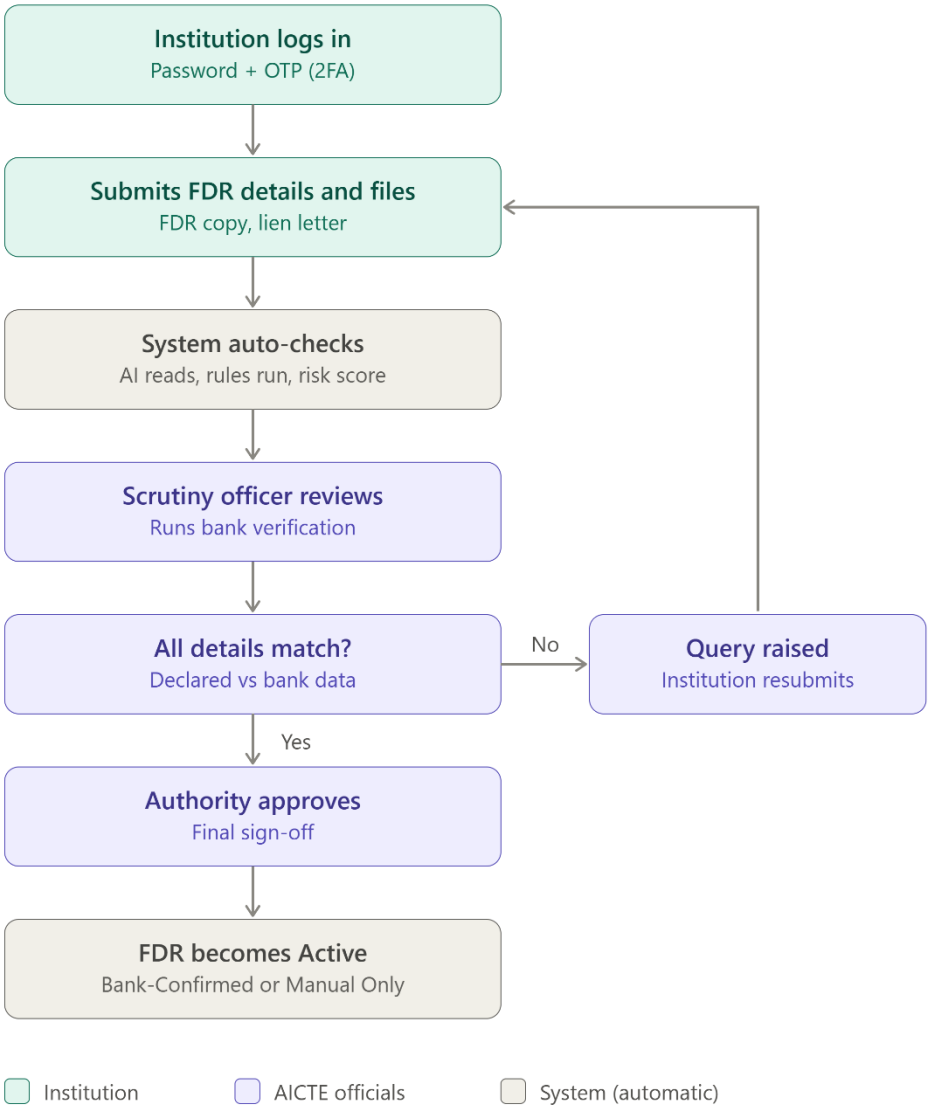
20	Secure access and protection of financial and institutional data	Role-based access (5 roles), OTP login, masked FDR and account numbers, a tamper-evident audit log, and AES-256 and TLS in the production design	RBAC; Audit; Data protection panel
21	Duplicate data entry across systems	An adapter layer is ready for the AICTE portal, NIC Parichay, DigiLocker, bank APIs and PFMS. These are planned or simulated, not live	Integration readiness page
22	Prototype must show measurable improvement	A before-and-after panel shows record accuracy, verification turnaround, on-time renewals, manual hours saved and visibility of pending cases. The figures are computed from seeded demo data, so they illustrate the effect rather than prove it	Impact metrics panel
23	Accountability and avoidable administrative delays	Every action is logged with who, what and when, and each approval stage has its own time limit	Audit trail; SLA timers

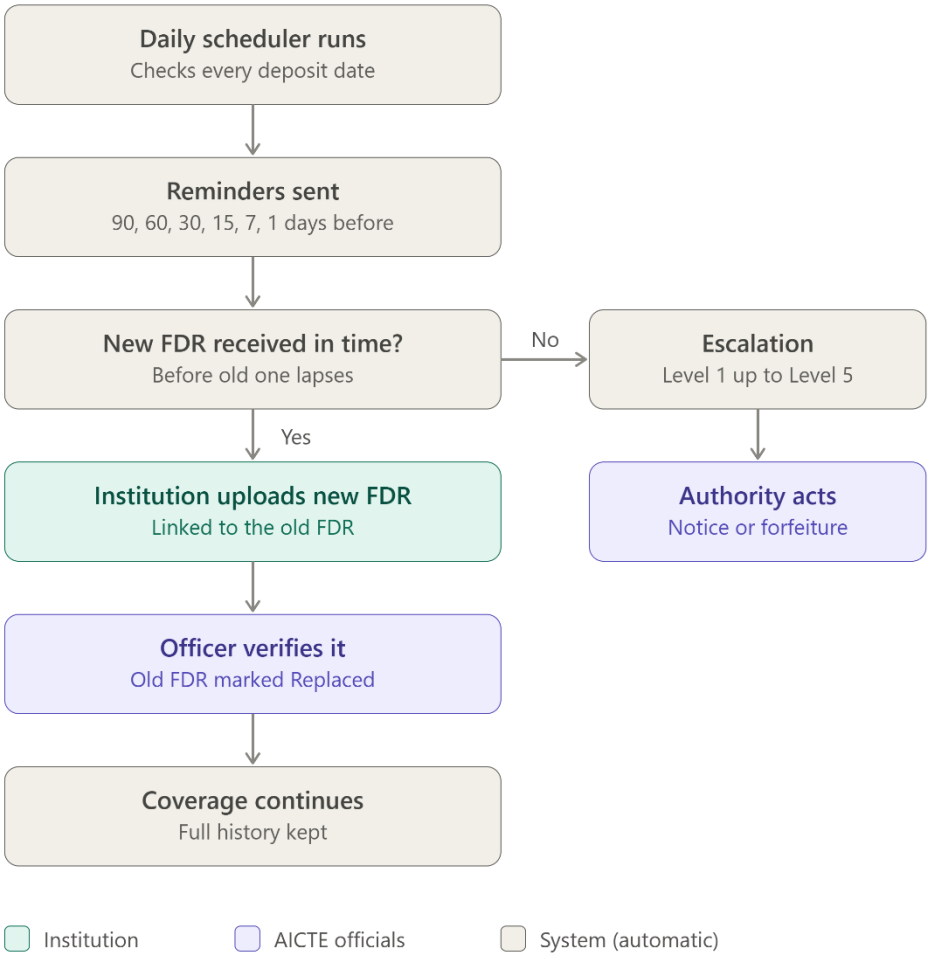
Testing scenarios in the statement

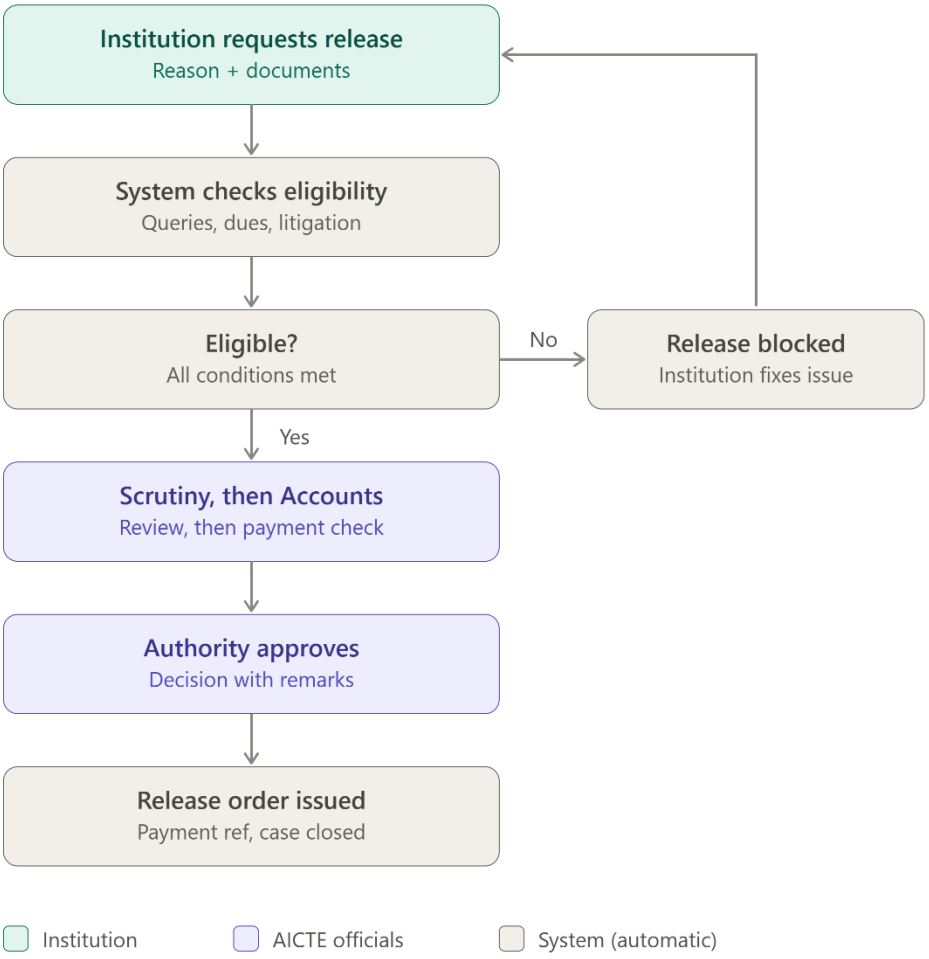
Scenario to test	How it is demonstrated
New FDR submission	Demo scenario 1: AI extraction followed by bank verification
Validity tracking	Demo scenario 2: an FDR with 28 days left and alerts firing
Approaching maturity	Demo scenario 3: an FDR with 12 days left, escalated to level 3
Renewal	Demo scenario 4: old FDR replaced and linked with no gap
Discrepancy identification	Demo scenario 6: amount mismatch, name mismatch and duplicate FDR
Authorized release processing	Demo scenario 7 (full approval to closure) and scenario 8 (release blocked)

The "Simulate time" button in the prototype lets you move the date forward during the demo, so reminders, status changes and escalations happen live.

Bank verification, SMS, email, e-Sign and the other external integrations are simulated in the prototype, so say that clearly when you present. The workflow, rules, role-based access, audit chain and AI document reading are real working features.


Institution logs in
Password + OTP (2FA)
Submits FDR details and files
FDR copy, lien letter
System auto-checks
AI reads, rules run, risk score
Scrutiny officer reviews
Runs bank verification
All details match?
Declared vs bank data
No
Query raised
Institution resubmits
Yes
Authority approves
Final sign-off
FDR becomes Active
Bank-Confirmed or Manual Only
Institution
AICTE officials
System (automatic)


Daily scheduler runs
Checks every deposit date
Reminders sent
90, 60, 30, 15, 7, 1 days before
New FDR received in time?
Before old one lapses
No
Escalation
Level 1 up to Level 5
Yes
Institution uploads new FDR
Linked to the old FDR
Authority acts
Notice or forfeiture
Officer verifies it
Old FDR marked Replaced
Coverage continues
Full history kept
Institution
AICTE officials
System (automatic)


Institution requests release
Reason + documents
System checks eligibility
Queries, dues, litigation
Eligible?
All conditions met
No
Release blocked
Institution fixes issue
Yes
Scrutiny, then Accounts
Review, then payment check
Authority approves
Decision with remarks
Release order issued
Payment ref, case closed
Institution
AICTE officials
System (automatic)

How each login works

1. Institution user (college nodal officer)

What they see: only their own college's FDRs, with a status colour and a "next action" on each.

1. Log in with password + OTP.
2. Click "Submit new FDR", then upload the FDR scan. The AI fills in the form and highlights doubtful fields in yellow.
3. Check the values, accept the declaration, e-sign, and get a printable receipt with a QR code.
4. If an officer raises a query, reply and upload the extra document.
5. When a reminder arrives, upload the renewal FDR before the old one expires.
6. When the approval cycle is over, raise a release request.

Cannot: see other colleges, approve anything, or change a verified record.

2. Scrutiny officer (AICTE Regulation Bureau)

What they see: the full dashboard, a "Needs attention today" list sorted by urgency, and every institution's records.

1. Open a new submission and compare three columns side by side: Declared, AI-extracted, Bank-confirmed. Mismatches show in red.
2. Click "Run bank verification". The result is Match, Mismatch, Unable to verify, or Not supported.
3. Raise an observation or ask for more documents if anything looks wrong.
4. Mark the instrument Verified and send it to the Authority.
5. Review release requests and renewals the same way.
6. Use "Ask the Portal" to search in plain English, and the AI-drafted letters (which must be confirmed before sending).

Cannot: give final approval or release money.

3. Approving authority (senior officer)

What they see: cases waiting for a decision, escalations, and critical-risk cases.

1. Open a pending case and read the officer's remarks and the risk explanation.
2. Approve or reject the verification.
3. Give final approval for release, refund or forfeiture.
4. Handle escalated cases such as a lapsed FDR with no replacement.

Cannot: edit documents or institution data.

4. Accounts / finance officer

What they see: only release and refund cases that have passed scrutiny.

1. Check the bank details and amount to be released.

2. Confirm the payment, enter the release or discharge reference number, and attach the proof.
3. Mark the case ready for closure.

Cannot: verify FDRs or approve release.

5. System admin / auditor

What they see: users, settings and logs. They are read-only for business data.

1. Create users and assign roles.
2. Set the reminder days, escalation timings, minimum deposit rules and document checklists.
3. Open the audit log, filter it, and press “Verify chain” to prove no record was altered.
4. Export logs and review access records.

Cannot: edit or approve any FDR.

Algorithms used, in simple words

Part of the system	Algorithm or technique	What it does in simple words
Login	Password hashing (Argon2 or bcrypt) + TOTP one-time code	Keeps passwords unreadable and adds a second check
Permissions	Role-based access control (RBAC)	Each role has a list of allowed actions; anything else is refused
Status control	Finite state machine	A status can only move along allowed steps, e.g. you cannot jump from Submitted to Released
Reading FDR scans	OCR + multimodal AI (Gemini) with confidence scores	Reads the document and says how sure it is about each field
Error checking	Rule engine (if-then rules)	Flags things like amount too low, maturity before issue date, or no lien to AICTE
Duplicate and name matching	Hash lookup + fuzzy matching (Levenshtein, Jaro-Winkler)	Finds the same FDR number used twice, or names that are almost the same
Risk score (0-100)	Weighted scoring in the prototype; gradient-boosted trees (XGBoost) with SHAP in production	Adds up risk points and shows which factors pushed the score up
Unusual patterns	Isolation Forest (optional)	Spots cases that look unlike normal ones
Alerts	Scheduled job (cron) + date arithmetic	Every night, subtracts today from each due date and sends reminders
Escalation	Timer-based escalation ladder	If nobody acts within the set days, the case moves up one level
Priority list	Weighted sort (risk x urgency)	Puts the most dangerous and most urgent cases on top
Ask the Portal	LLM turns the question into a JSON filter, and the app runs it on the database	The AI only writes the filter, so it cannot invent data

Forecast	Aggregation + moving average	Counts upcoming maturities month by month and projects the workload
Tamper-proof audit	SHA-256 hash chain	Each log entry includes the fingerprint of the previous one, so any edit breaks the chain
Document integrity	SHA-256 file hash	Shows whether an uploaded file was changed later
Data protection	AES-256 at rest, TLS 1.3 in transit	Stored and transferred data is encrypted

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Integrations needed

Integration	Purpose	Status
Bank verification (bank API / core banking confirmation)	Confirm that the FDR is real, with the right amount, dates and lien	Simulated in the prototype; needs bank agreements
AICTE approval portal	Fetch institution details and approval data, avoid re-typing	Planned
NIC Parichay (SSO)	Government single sign-on for officers	Planned
DigiLocker	Fetch documents directly from the institution	Planned
e-Sign (Aadhaar-based or DSC)	Legally valid digital signature on declarations and orders	Simulated
SMS and email gateway (NIC)	Send reminders and escalations	Simulated
PFMS / Bharatkosh	Process refund payments	Planned
Gemini API (prototype) or an in-country hosted model (production)	Document reading, drafting, natural-language search	Real in the prototype
MeghRaj / NIC cloud	Hosting inside India	Production

All of these sit behind an "adapter" layer, so a simulated service can later be replaced by the real one without rewriting the app.

Programming languages and tools

Layer	Prototype in AI Studio	Production version
Front end	TypeScript, React, Tailwind CSS, Recharts	Same (TypeScript + React)
Back end	A simulated service layer in TypeScript with browser storage	Java (Spring Boot) or Python (FastAPI), exposed as REST APIs
AI and data science	Gemini API called from TypeScript	Python (scikit-learn, XGBoost, SHAP)
Database	localStorage	PostgreSQL (SQL); Redis for caching
File storage	Browser	Encrypted object storage (MinIO or S3-compatible)
Scheduler and alerts	A "simulate time" button in JavaScript	Cron or Celery with RabbitMQ or Kafka
Search	Filters in JavaScript	PostgreSQL full-text or Elasticsearch
Deployment	AI Studio preview	Docker + Kubernetes on MeghRaj/NIC